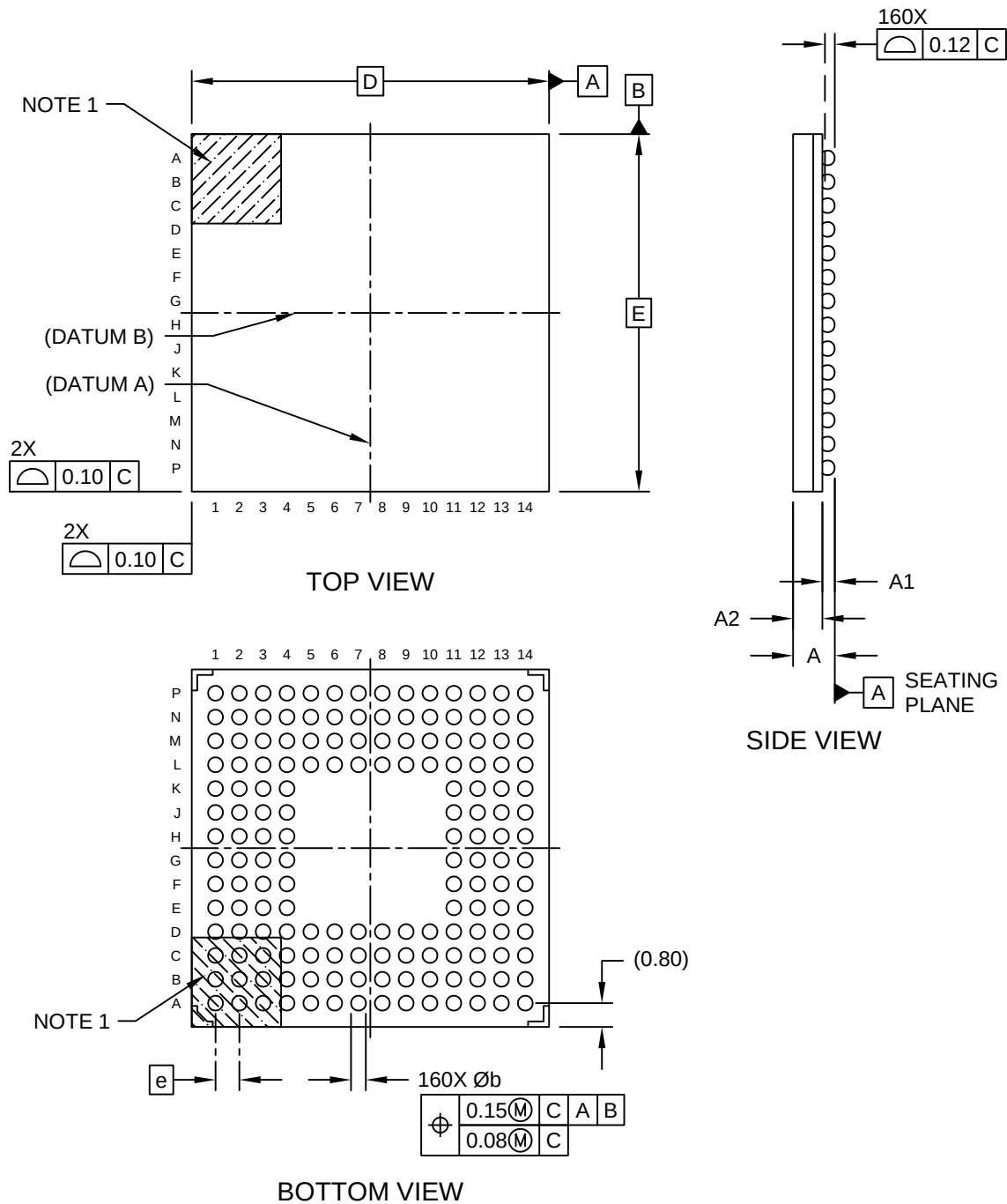


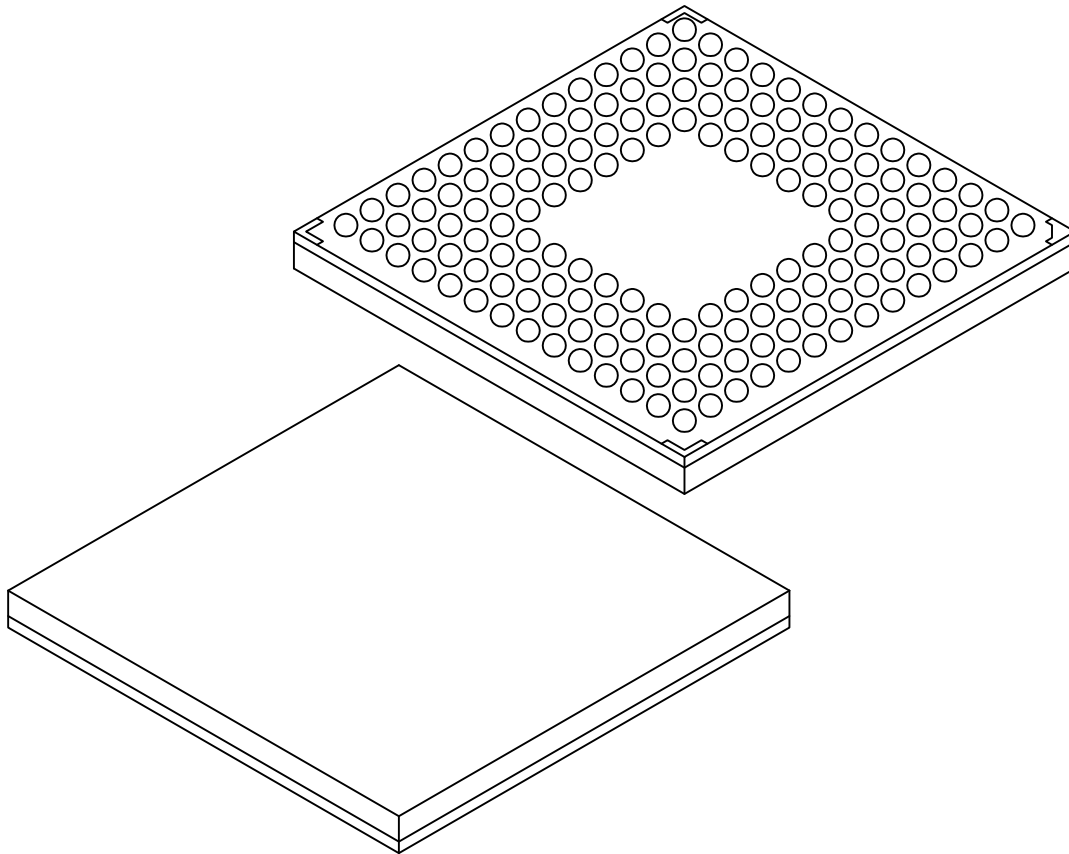
160-Ball Low Profile Fine Pitch Ball Grid Array (ADB) - 12x12x1.4 mm [LFBGA]
Atmel Legacy Global Package Code CFB

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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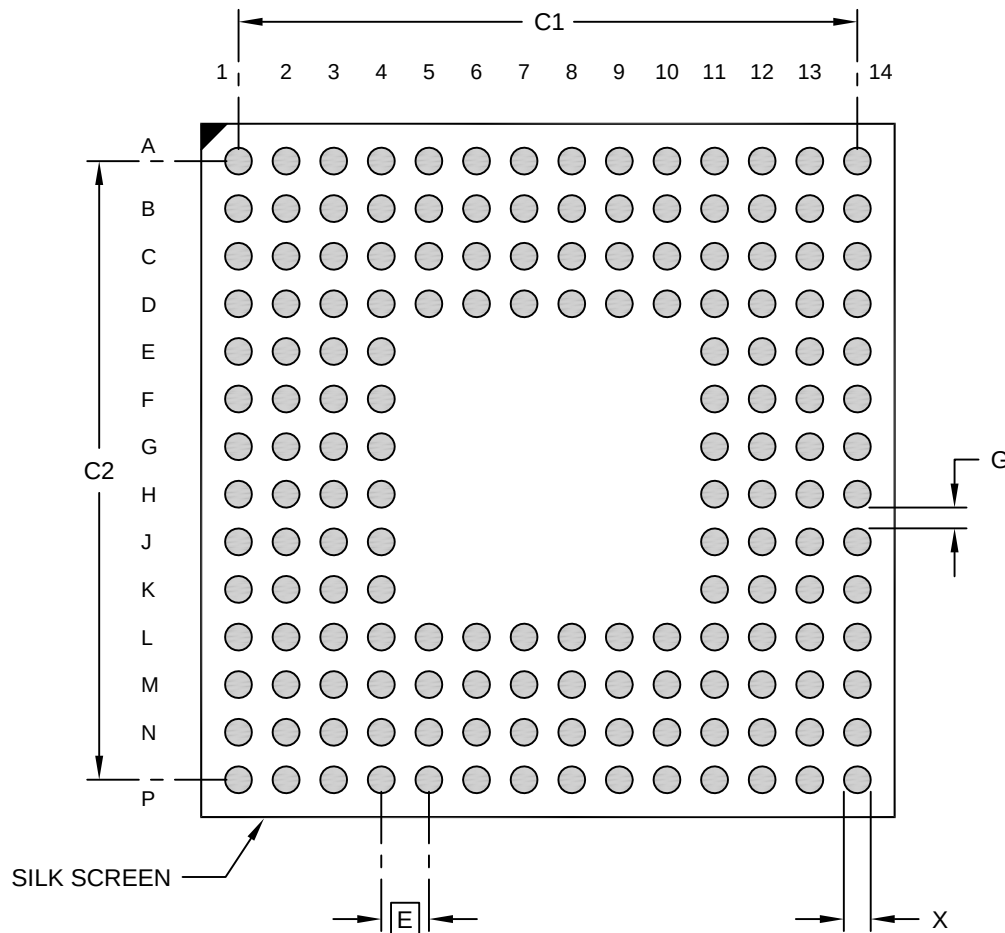
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	160		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.40
Ball Height	A1	0.25	-	-
Package Thickness	A2	0.85	-	-
Overall Length	D	12.00 BSC		
Overall Width	E	12.00 BSC		
Terminal Width	b	0.45	0.50	0.55

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
- BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

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RECOMMENDED LAND PATTERN

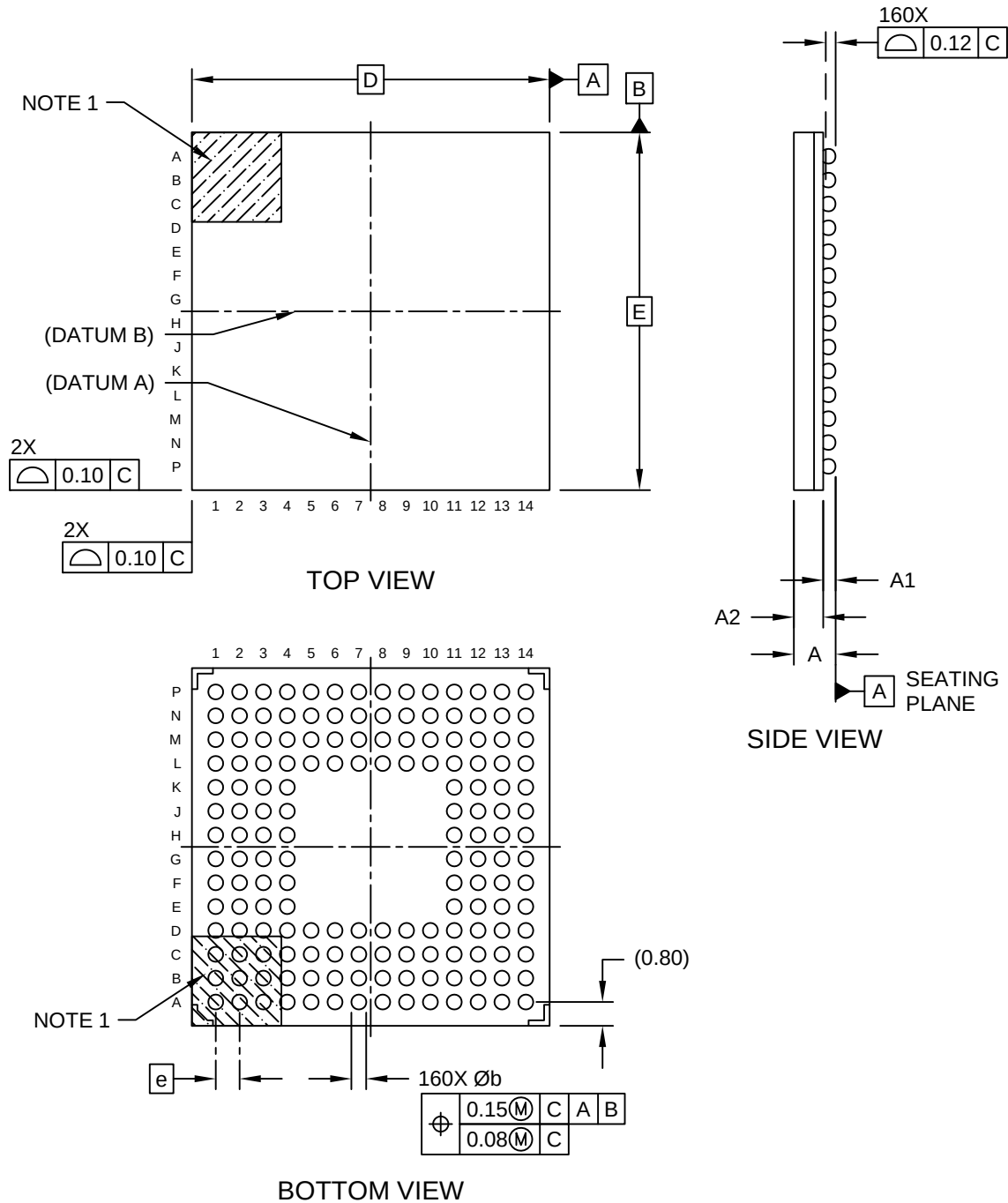
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1		10.40	
Contact Pad Spacing	C2		10.40	
Contact Pad Width (X160)	X			0.45
Contact Pad to Contact Pad	G	0.35		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

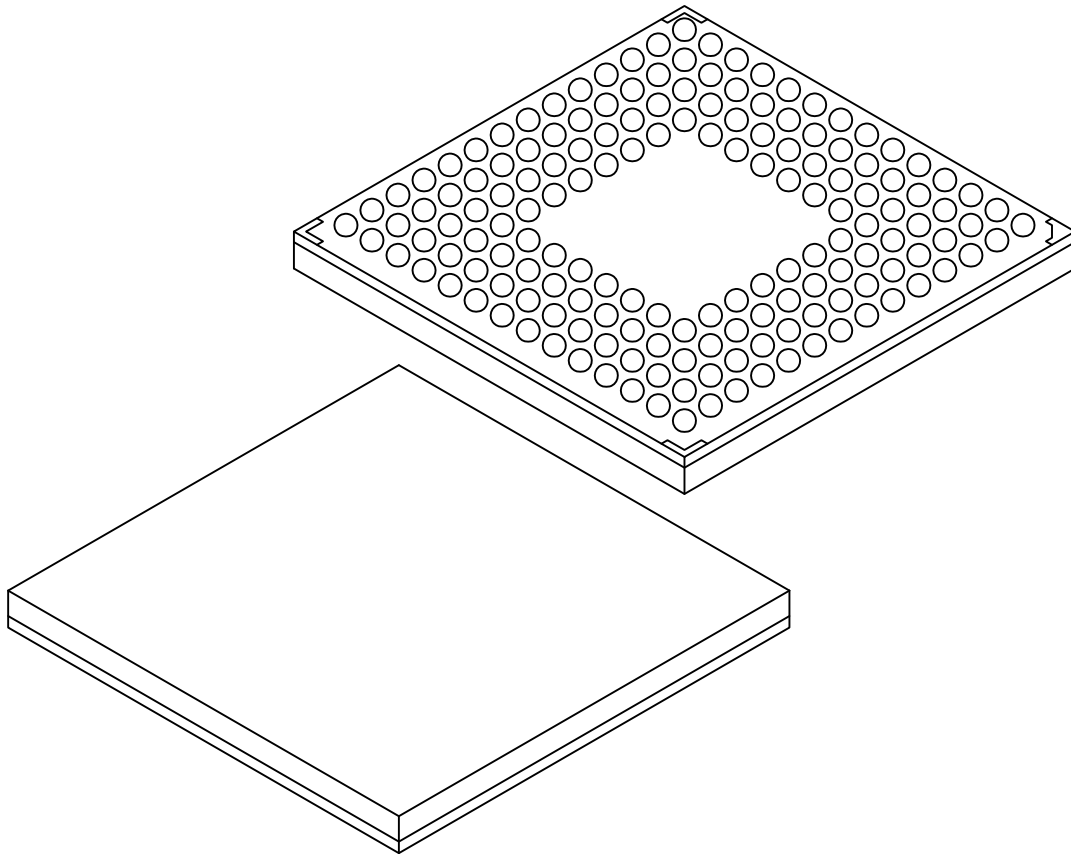
160-Ball Low Profile Fine Pitch Ball Grid Array (AEB) - 12x12x1.4 mm [LFBGA] **Atmel Legacy Global Package Code CFB**

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**160-Ball Low Profile Fine Pitch Ball Grid Array (AEB) - 12x12x1.4 mm [LFBGA]
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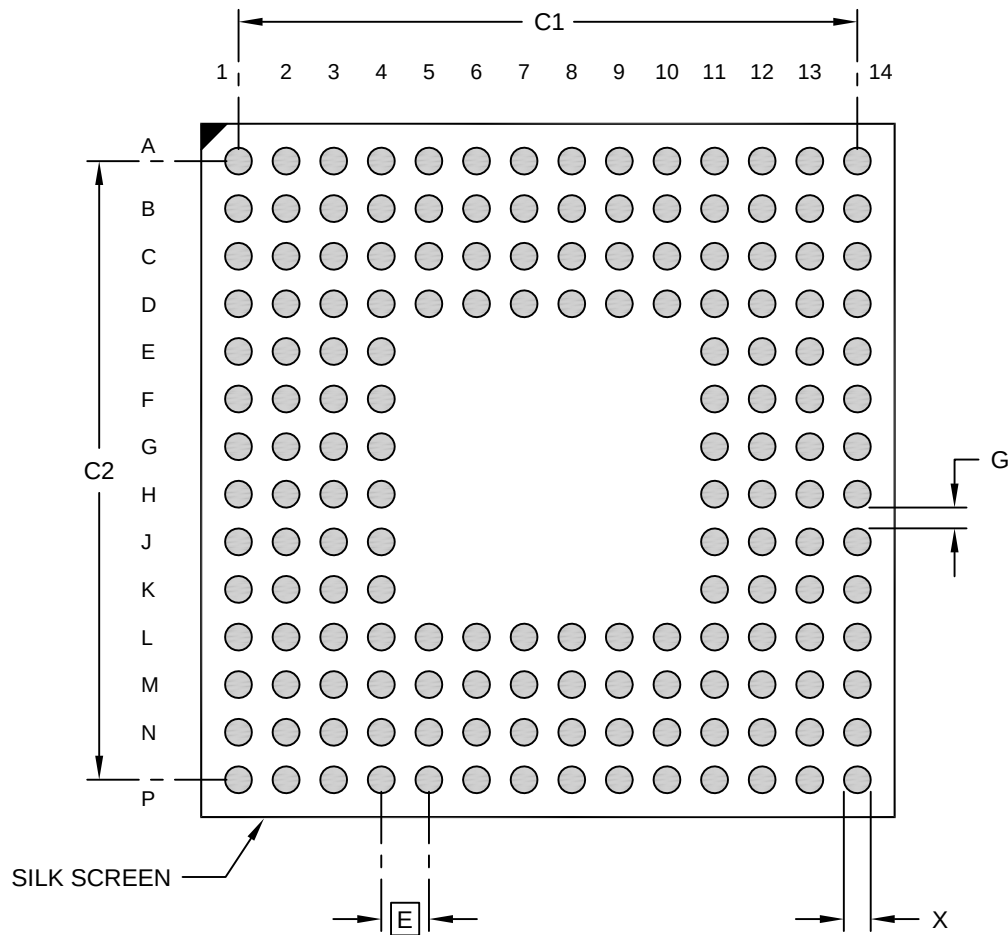
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	160		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.40
Ball Height	A1	0.25	-	-
Package Thickness	A2	0.85	-	-
Overall Length	D	12.00 BSC		
Overall Width	E	12.00 BSC		
Terminal Width	b	0.45	0.50	0.55

Notes:

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RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1		10.40	
Contact Pad Spacing	C2		10.40	
Contact Pad Width (X160)	X			0.45
Contact Pad to Contact Pad	G	0.35		

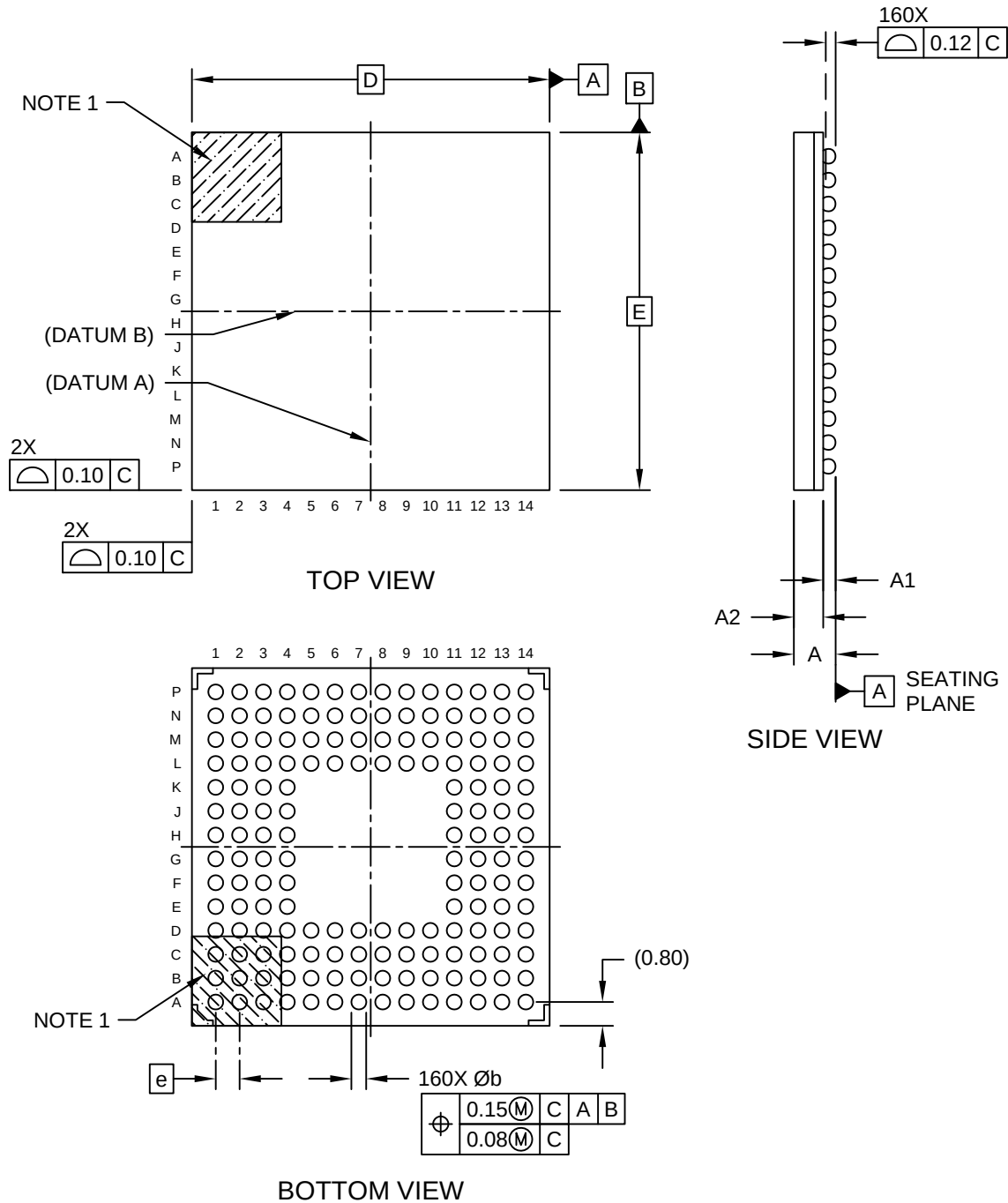
Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

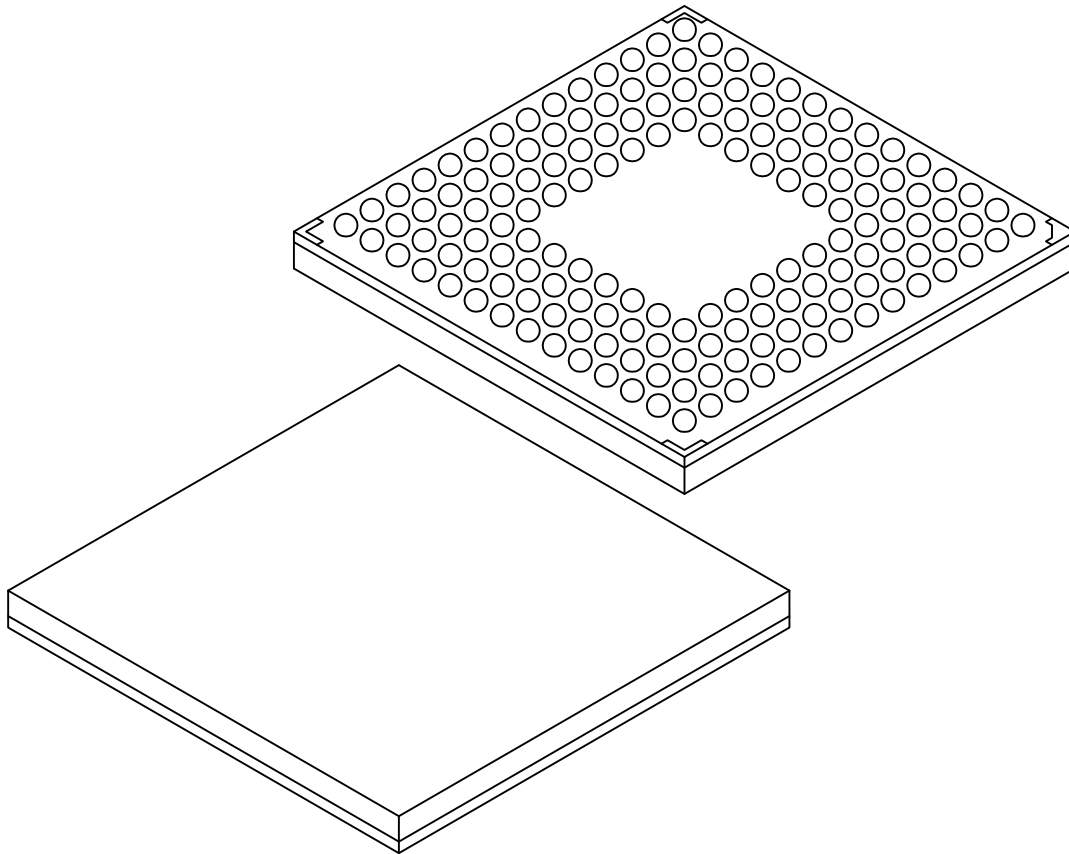
160-Ball Low Profile Fine Pitch Ball Grid Array (AFB) - 12x12x1.4 mm [LFBGA] **Atmel Legacy Global Package Code CFB**

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160-Ball Low Profile Fine Pitch Ball Grid Array (AFB) - 12x12x1.4 mm [LFBGA]
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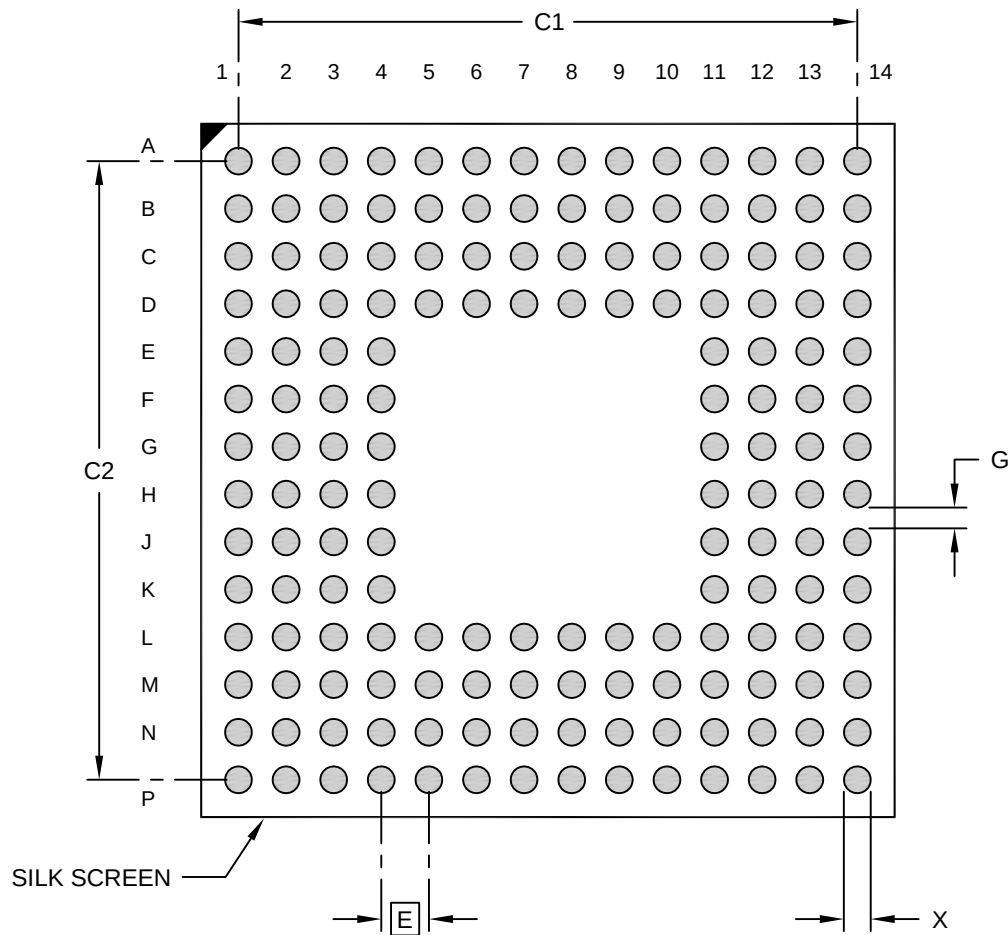
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	160		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.40
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